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Grade
Kacharam, Shamshabad, Hyderabad- 501 218,

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

III B. TECH II SEMESTER

BATCH 2023-27	MINI PROJECT 2025-26	ACADEMIC YEAR 2025-26

Batch Id : 23MPCS-H5
Title of the Project : **Medicine Overdose Prediction**

TEAM MEMBERS

S. No.	Roll Number	Student Name	Mail Id	Contact Number
1	23881A05GX	Siri Manvitha Sheela	sirimanvitha504@gmail.com	8341548099
2	23881A05ER	Akula Rishi Preetham	rishiakula70@gmail.com	7013291809
3	23881A05GW	Siddamsetty Vishwen	vishwen2005@gmail.com	8688031930

GUIDE DETAILS

Name of the Guide	:	**Dr. Gouse Baig Mohammad**
Designation	:	Dean outreach & Associate professor
Department	:	Computer Science and Engineering
Mail Id	:	Gousebaig@vardhaman.org
Contact Number	:	9959593404
Area of interest	:	Network security, Iot , Blockcahin technology

Signature of the Projects In charge

Signature of the Guide with Date

Signature of the Project Coordinator

Batch Id **:** 23MPCS-H5

Title of the Project **: Medicine Overdose Prediction**

Objectives

1. To build a system that predicts the chances of medicine overdose using patient data
2. To collect and analyze important details such as patient history, age, dosage, and medicine combinations.
3. To apply machine learning models (like decision trees or neural networks) to find patterns that may lead to overdose.
4. To integrate the system with electronic health records for real-time alerts and safer medical decisions.
5. To help doctors and pharmacists by giving early warnings about risky prescriptions.

Outcomes

1. A working model that can accurately predict medicine overdose risks.
2. Early alerts for healthcare professionals to prevent harmful drug interactions.
3. Reduced medication-related hospitalizations and improved patient safety.
4. A smart decision-support tool for clinicians during prescription.
5. Better awareness and management of dosage and drug combinations through data-driven insights.

ABSTRACT

Medicine overdose prediction helps prevent harmful drug reactions and accidental overdoses by analysing patient data like age, medical history, prescribed drugs, dosage, and timing. Using machine learning methods such as decision trees, random forests, and neural networks, the system detects hidden patterns to warn of potential overdose risks. It supports doctors and pharmacists by flagging dangerous drug combinations, suggesting safer dosages, and recommending monitoring steps. This improves patient safety, reduces hospital visits, and aids clinical decisions. Success depends on accurate data, good feature selection (e.g., drug interactions, organ health), and medical expertise. Integration with electronic health records and real-time alerts enhances its use in healthcare settings.

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Signature of the Guide with Date

Signature of the HOD, CSE